

Naive-Phantom Rate Stability

AI Presence Index v0.10 — Mint, designed-for-test extension,
 $t_1 \rightarrow t_2$

9 May 2026 · Pablo Ulpiano González Castro · Third System™

Independent measurement for the AI mediation layer.

The phantom is mostly extinguished. Mostly.

v0.7 measured a small but non-zero naive-phantom rate for Mint at the v0.6 baseline (1.7 percent). v0.9 confirmed gross Presence stability at matched-subset n. v0.10 tests the question v0.9 H4 left open: is the naive-phantom subset — recommendation slots delivered without caveat — as durable as gross Presence? The matched-subset n breached the pre-registered floor and routed H1 to descriptive disclosure. Within that path: zero naive at t_1 (43 of 43 caveated), one naive at t_2 (1 of 40). H2 falsified. H3 decoupled. The single t_2 case stands as existence proof that naive-phantom presence has not yet been fully extinguished from frontier-model substrates.

Third System applied the v0.7 caveat-classifier to v0.9's deposited raw responses, restricted to Mint within the matched two-model subset (Anthropic Claude Sonnet 4.6 and OpenAI gpt-5.4-mini). Three hypotheses were pre-registered:

H1 stability (± 2 pp band, $n \geq 100$ floor), **H2 persistence** ($r_{\text{naive}} > 0$ at both waves), and **H3 co-movement** (diagnostic).

Pre-registration locked at git commit **8767f44**, tag

v0.10-prereg, on 2026-05-09, prior to any application of the classifier to the Mint subset.

The floor breached. Matched-subset n at $t_1 = 43$; at $t_2 = 40$. Both below the pre-registered $n \geq 100$ required for H1 confirmatory inference. Pre-reg §3.4 routed H1 to the descriptive disclosure path — exactly the case the floor was specified to handle. The pre-registered band is not invoked; the small-n descriptive picture is reported in its place.

H2 falsified. $r_{\text{naive}, t_1} = 0.00$ percent (0/43); $r_{\text{naive}, t_2} = 2.50$ percent (1/40). Every Mint mention at t_1 in the matched subset carried some form of caveat — 100 percent classifier-detectable correction at the v0.6 baseline measurement window. The single naive case at t_2 is in claude-sonnet-4-6, prompt **p1_functional**, run 3: Mint listed at the top of “Free Options”, described in present tense with feature details, and explicitly recommended as “the most common starting point for beginners.” No caveat. Adjudication confirmed the classifier without override.

H3 decoupled. Gross Presence falls (-3.12 pp from 44.79 to 41.67 percent) while the naive rate rises ($+2.50$ pp from 0 to 2.5 percent). The directional change is opposite. At small n the pattern is noise-dominated and reported as diagnostic only. If substantively replicated at higher n, the asymmetry would suggest the naive subset and the gross subset are governed by partly different mechanisms.

The result is honest, narrow, and methodologically clean.

The pre-registered floor caught a real small-n problem before it could be rationalized away. Four deviations from the pre-registration's descriptive language — the v0.7 classifier is LLM-only, not the rule-list-plus-judge protocol pre-reg §5 described — are documented in DEVIATIONS.md and summarized in the SSRN paper. The substantive instruction (“inherit v0.7 unchanged”) is preserved. v0.10 ships as descriptive disclosure plus existence proof rather than confirmatory measurement.

What we measured

v0.7 (Phantom-Brand BBB) introduced a designed-for-test reframe distinguishing two phantom-brand presence types: **naive-phantom presence** (recommendation as live, no caveat anywhere in the response) and **caveated-phantom presence** (mention with explicit acknowledgment of decommissioning, sunset, or migration). At the v0.7 BBB measurement against the v0.6 baseline data, the naive-phantom rate for Mint was 1.7 percent, against a gross Presence rate of 38.2 percent.

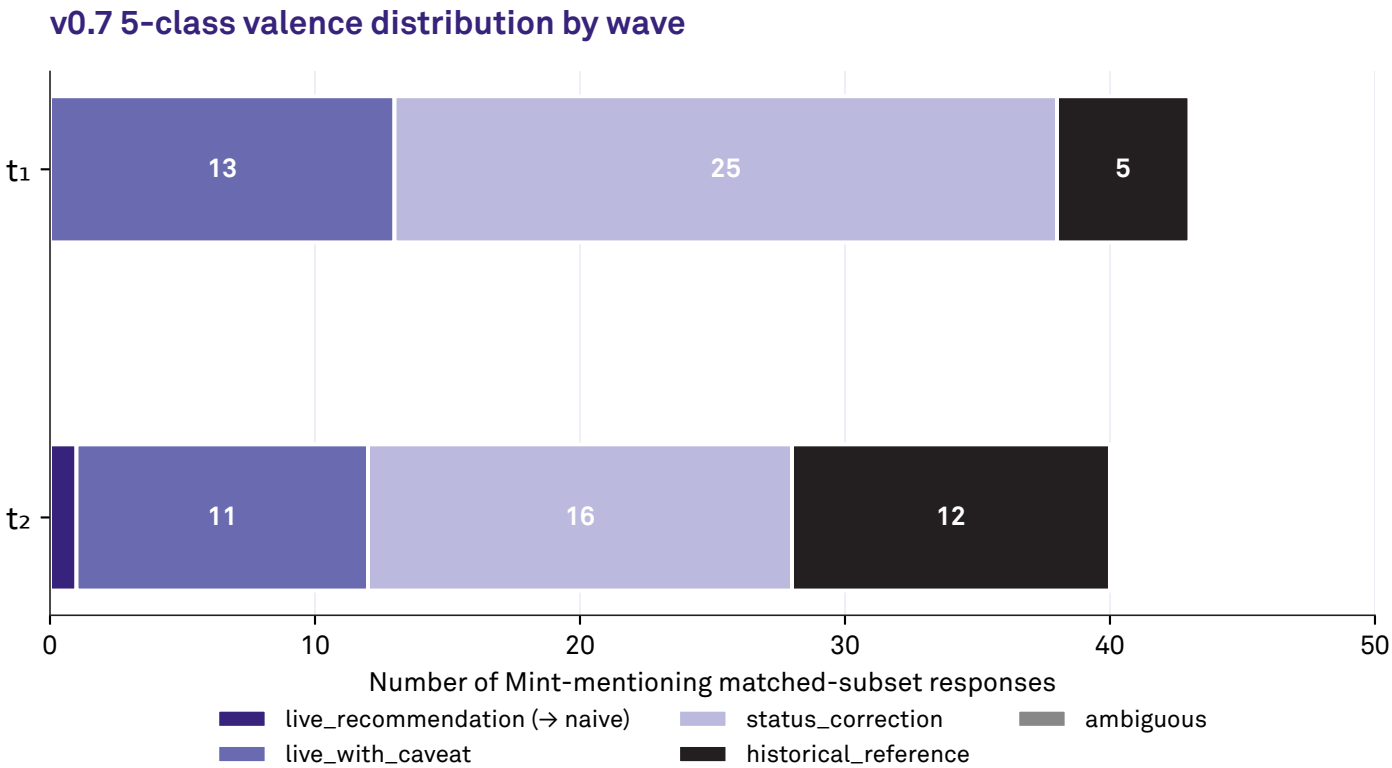
v0.9 (Longitudinal Re-Baseline) confirmed gross Presence stability for Mint across the v0.6→v0.9 measurement interval within the pre-registered ± 5 pp band (44.8 percent → 41.7 percent, $|\Delta| = 3.1$ pp). What v0.9 H4 did **not** test is whether the naive-phantom subset exhibits comparable stability — whether AI systems' tendency to recommend Mint as if fully live persists at the same rate over the longitudinal interval, separately from the gross-mention rate.

v0.10 tests this question. The v0.7 caveat-classifier is applied to the Mint-restricted subset of v0.9's deposited raw responses across the matched two-model subset (Sonnet 4.6, gpt-5.4-mini). No new measurement; no new model calls; no new data collection. The pre-registration locks three hypotheses with explicit numerical thresholds and an effective-n floor of 100 per wave. Floor breach routes H1 to descriptive disclosure as pre-specified.

All raw responses, classifier outputs, scoring scripts, adjudication log, and four deviations from pre-reg descriptive language are deposited at OSF project **ec6wh**, path **/v10/**. The pre-registration document is locked at git commit **8767f44**, tag **v0.10-prereg**, 2026-05-09.

FINDING 01

The floor breached. Descriptive disclosure engages.



Source: AIAS Presence Index v0.10 · Third System™. Mint × matched-subset (Sonnet 4.6, gpt-5.4-mini). n = 83 (t₁ = 43; t₂ = 40).

Figure 1 · Distribution of the v0.7 5-class valence output across both waves before mapping to the v0.10 binary. *status_correction* (the strongest caveat type — explicit “Mint is decommissioned/closed/sunset” framing) dominates both waves. *historical_reference* and *live_with_caveat* make up the remainder of caveated cases. Zero *ambiguous* classifications were produced; the E3 sensitivity mapping is therefore identical to the primary mapping.

Exact-token match of *Mint* in the v0.9 raw responses’ *brands_canonical* field, restricted to the matched two-model subset (Sonnet 4.6 + gpt-5.4-mini), yielded **43 Mint-bearing rows at t₁** and **40 at t₂**. Both well below the pre-registered n=100 floor for H1 confirmatory inference.

Pre-reg §3.4 specified the floor as a single-flip resolution constraint: at n=100, a single response classification change moves r_{naive} by 1.0pp — half the ±2pp stability band. Below that floor, the test cannot distinguish meaningful drift from sampling noise. The specified response is to route H1 to descriptive disclosure, not to relax the threshold or promote the descriptive observation to a confirmatory claim.

The classifier output — v0.7’s 5-class valence (*live_recommendation*, *live_with_caveat*, *status_correction*,

historical_reference, *ambiguous*) — is dominated by **status_correction** at both waves: 21 of 43 at t₁ (48.8 percent) and 20 of 40 at t₂ (50.0 percent). This is the strongest caveat type, an explicit “Mint is decommissioned/sunset/closed” framing, distinct from a hedged recommendation.

historical_reference and *live_with_caveat* make up the remainder of the caveated cases. Zero *ambiguous* classifications were produced; the E3 sensitivity mapping (which would re-route ambiguous cases to the caveated bin) is therefore identical to the primary mapping.

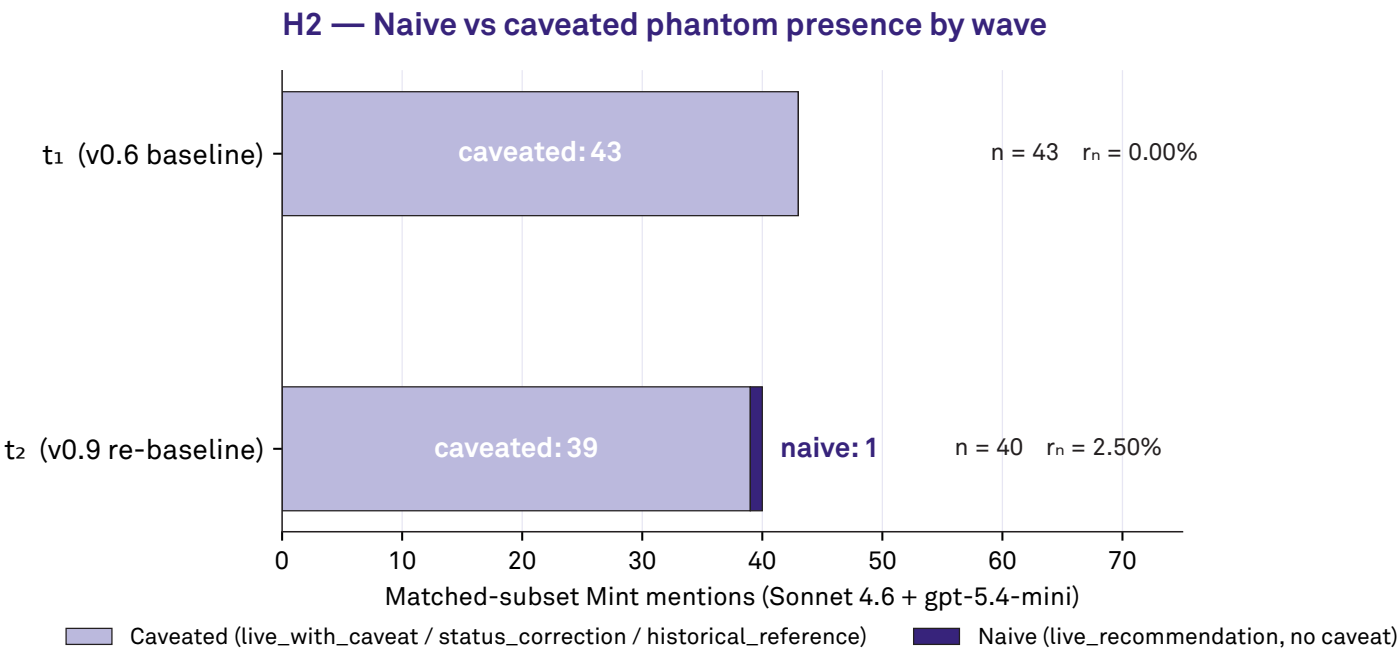
The descriptive picture is clean even under the floor breach. Within the data we have, classifier-detectable correction covers virtually every Mint surfacing in the matched-subset frontier-model substrate — and at t₁ specifically, every single

one. The proper read is not “measurement failed”; it is “measurement honestly disclosed below confirmatory

threshold,” with three pre-registered observations to follow.

FINDING 02

H2 falsified. One Sonnet 4.6 case is existence proof.



Source: AIAS Presence Index v0.10 · Third System™. Mint × matched-subset (Sonnet 4.6, gpt-5.4-mini). n = 83 (t₁ = 43; t₂ = 40).

Figure 2 · Per-wave naive vs caveated phantom-presence breakdown for the matched two-model subset. At t₁, all 43 Mint mentions carried some form of caveat; zero were naive. At t₂, 39 of 40 carried caveats and one was naive. The single naive case is a Sonnet 4.6 response listing Mint at the top of “Free Options” with present-tense feature description and explicit recommendation as “the most common starting point for beginners.” No caveat anywhere in the response.

H2 (persistence): $r_{naive} > 0$ at both waves. **Falsified.** $n_{naive,t_1} = 0$; $n_{naive,t_2} = 1$. Every Mint mention at t₁ in the matched subset carries some form of caveat — 100 percent classifier-detectable correction at the v0.6 baseline measurement window.

This is a stronger correction baseline than v0.7 BBB recorded for Mint at the same v0.6 measurement (which found 1.7 percent naive). Three non-mutually-exclusive explanations: v0.7’s classifier was applied to a broader response set including models beyond the matched two; substrate evolution between v0.7’s measurement window (4 May 2026) and v0.10’s (9 May 2026, applied to v0.9’s t₁ collection); and small-n sampling noise within the matched-subset restriction. The data do not adjudicate among them.

The single t₂ naive case is claude-sonnet-4-6, prompt p1_functional, run 3. The response lists Mint at the top of a “Free Options” list. It describes the product’s features in present tense — “Comprehensive, links to bank accounts, automatic categorization” — and explicitly recommends it as “the most common starting point for beginners.” No caveat about decommissioning, shutdown, or migration to Credit Karma anywhere in the response. The classifier valence (*live_recommendation*) and binary mapping (naive) were both confirmed without author override.

The case constitutes existence proof at the substrate level. A frontier model (Sonnet 4.6) recommends a decommissioned product — Mint, two years past decommissioning at the t₂ collection date — as a top current option in its category, with present-tense feature description and explicit endorsement

as a beginner's first choice. The phenomenon is not yet fully extinguished from frontier-model retrieval and response-generation pipelines.

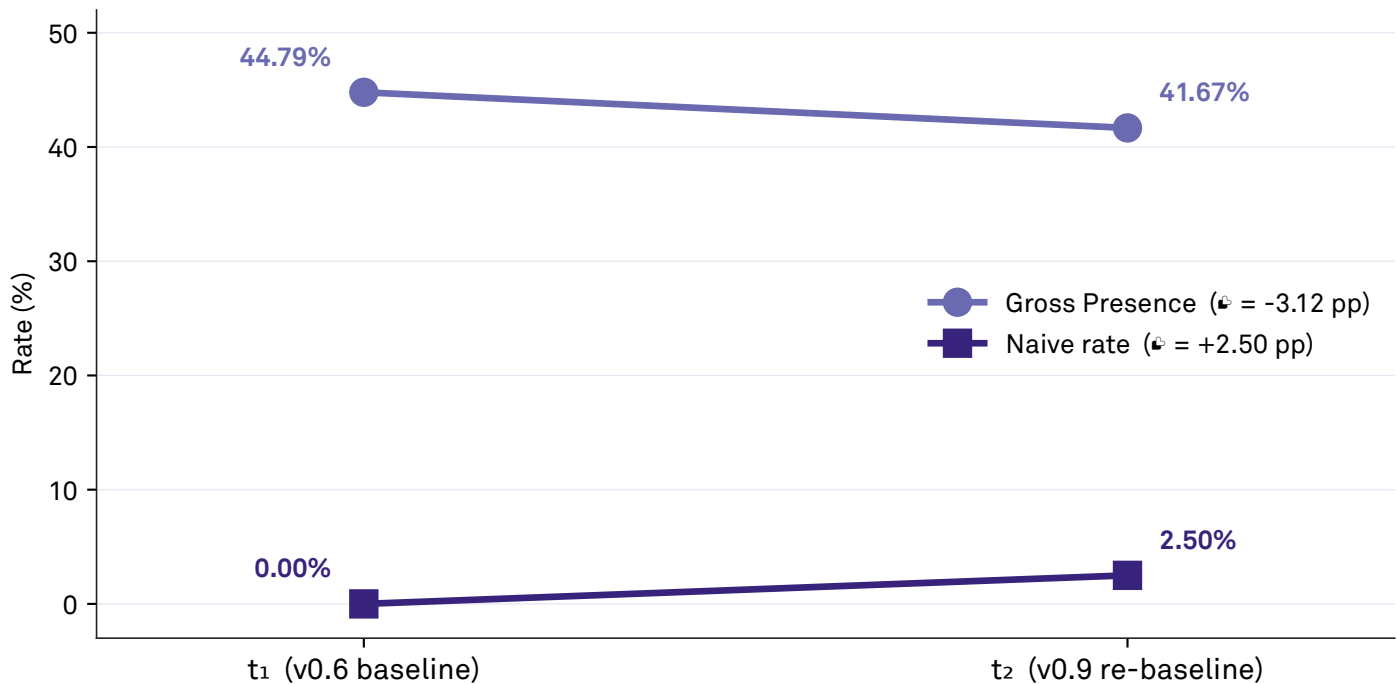
What the case is **not**: representative. One naive instance in 40 is consistent with rates anywhere from essentially zero to a

few percent at population scale. What v0.10 has shown is that the rate is non-zero in this wave and zero in the previous wave — enough to falsify H2's formal prediction ($r_{\text{naive}} > 0$ at **both** waves) without resolving the question of whether naive-phantom presence is structurally drifting in either direction.

FINDING 03

H3 decoupled. Gross down, naive up.

H3 — Directional decoupling: gross down, naive up



Source: AIAS Presence Index v0.10 · Third System™. Mint × matched-subset (Sonnet 4.6, gpt-5.4-mini). $n = 83$ ($t_1 = 43$; $t_2 = 40$).

Figure 3 · Directional decoupling of gross Presence and naive-phantom rate across $t_1 \rightarrow t_2$. Gross Presence falls from 44.79% to 41.67% ($\Delta = -3.12\text{pp}$) while the naive rate rises from 0.00% to 2.50% ($\Delta = +2.50\text{pp}$). The directions are opposite; the magnitudes are comparable in absolute terms. At the matched-subset n observed (43 and 40), the pattern is noise-dominated and reported as diagnostic only per pre-reg §2 H3.

H3 (co-movement, diagnostic): the directional change in naive-phantom rate is consistent with the directional change in gross Presence. **Decoupled.** $\Delta_{\text{gross}} = -3.12\text{pp}$; $\Delta_{\text{naive}} = +2.50\text{pp}$. The directions are opposite; the magnitude ratio $|\Delta r_{\text{naive}}| / |\Delta r_{\text{gross}}| = 0.80$.

At the matched-subset n observed (43 and 40), the pattern is noise-dominated. A single classification flip in either wave moves r_{naive} by 2.3 to 2.5pp. The H3 result is reported as diagnostic only per pre-reg §2; no falsification status is

assigned, regardless of how striking the visual impression of decoupling looks.

If substantively replicated at higher n , the asymmetry would be informative. Two readings, not mutually exclusive: **(a)** as caveat coverage matures across the model substrate, the residual naive cases become structurally harder to correct — perhaps because they arise from prompt formulations or response paths that bypass the typical correction pipeline. **(b)** the naive subset and the gross subset are governed by partly

different mechanisms: gross presence by training-data prevalence and retrieval propensity, naive subset by the response-time decision to caveat or not.

Either reading would have implications for measurement strategy. The first suggests naive-phantom rates may stabilize near a positive floor rather than tend to zero. The second suggests gross Presence and naive-phantom rate should be

tracked as separate metrics with separate stability bands rather than treated as proportional or co-moving.

A higher-volume study at $n \approx 100$ per wave would distinguish noise from signal. The same dataset cannot. v0.10 records the diagnostic for the program log; confirmatory inference about co-movement waits.

Limitations

Sample size. The matched-subset n at both waves is below the pre-registered floor of 100 per wave. Descriptive observations only; no inference about the population-level naive-phantom rate beyond the v0.6→v0.9 longitudinal window measured here.

Single-brand scope. v0.10 measures Mint only. Generalization to other phantom brands requires additional study. The v0.7 BBB result (1.7 percent naive at t_1) and v0.10 Mint result (0 percent at t_1 , 2.5 percent at t_2), taken together, suggest naive rates in the low-single-digit percentage range — consistent across two brands, but a sample of two.

Classifier validation. The v0.7 classifier is a single gpt-5.4-mini call at temperature 0 with the v0.7 system prompt and tool schema applied. It has not been independently calibrated for inter-rater agreement at the v0.10 measurement window. The single naive case was author-adjudicated and confirmed; no override was applied. Author agreement with the classifier on the high-information case (the single naive instance) is not equivalent to calibration on the broader caveated set.

Substrate evolution. AI training and retrieval substrates evolve continuously. The present result is a snapshot within v0.9's measurement windows (April-May 2026); it does not characterize the phenomenon's trajectory across longer time scales or across documented model-pipeline transitions.

Pre-reg description vs v0.7 reality. Four deviations from the pre-registration's descriptive language are documented in DEVIATIONS.md. The substantive pre-reg instruction ("inherit v0.7 unchanged") is preserved. Deviations stem from a single root cause: pre-reg §5 described the v0.7 classifier as a "keyword/phrase rule list and judge-LLM verification protocol"; the actual v0.7 classifier is LLM-only with no rule list and no separate judge step. The classifier configuration applied at v0.10 is the actual v0.7, documented at registries/v07_caveat_classifier.json.

What's next

Higher-volume replication. A v0.10b collection at a third time point t_3 specifically scaled to clear the $n \geq 100$ floor would enable confirmatory inference on H1. The most direct design is higher prompt-count per wave on the same matched subset, or careful expansion of the matched subset to include additional models where methodologically defensible (the matched-subset constraint exists to control for cross-model heterogeneity; relaxing it for n requires explicit accounting).

Multi-brand replication. Extending the v0.7/v0.10 protocol to additional phantom brands — Toys “R” Us, Circuit City, RadioShack, Pier 1, JCPenney post-bankruptcy, or comparable domain-specific phantoms — would establish whether the low-single-digit naive rate is a category-general property of the phenomenon or specific to particular brands. A designed-for-test multi-brand naive-phantom study is the natural v0.11 candidate.

Longer-interval longitudinal characterization. The present seven-day $t_1 \rightarrow t_2$ interval is short relative to typical model retraining cycles. Repeating the v0.10 protocol at one-month, three-month, and twelve-month intervals — particularly across documented model-pipeline transitions or training-cutoff updates — would characterize the phenomenon’s trajectory at scales relevant to substrate evolution rather than within-window noise.

Mechanism-level decomposition. If the H3 decoupling diagnostic replicates at higher n , the naive subset and gross subset may be governed by partly different mechanisms. Designed-for-test studies isolating the response-time decision to caveat (as opposed to the retrieval-time decision to surface) would clarify the mechanism. Candidates: prompt-level cues that force or suppress temporal context; ablation of specific portions of the response generation context; comparison of decoder-time behavior at matched retrieval states.

Methodology paper. A v1.x AIAS Presence Measurement Protocol paper, building on v1.1 and the v0.7→v0.10 designed-for-test thread, will document the naive-vs-caveated distinction as a first-class dimension of the measurement program rather than a v0.7-specific extension. v0.10’s descriptive disclosure path — and the four deviations between pre-reg description and v0.7 classifier reality — are useful inputs.

Pre-registration outcome

Three hypotheses were pre-registered. H1 was routed to descriptive disclosure under the n=100 floor breach. H2 was evaluated as confirmatory and falsified. H3 was reported as diagnostic without falsification status. Detailed evaluation, per-wave metrics, and the single naive case quote in full are in the v0.10 SSRN paper (deposited at OSF project ec6wh, /v10/papers/).

H	Pre-registered prediction	Result	Status
H1	$ \Delta r_{naive} \leq 2.0pp$; n=100 floor required at both waves	Floor breached: $n_{t_1} = 43$; $n_{t_2} = 40$. Routed to descriptive disclosure path per pre-reg §3.4. Descriptive $ \Delta r_{naive} = 2.50pp$.	INDETERMINATE
H2	$r_{naive} > 0$ at both waves	$n_{naive,t_1} = 0$; $n_{naive,t_2} = 1$. The single naive case is sonnet-4-6, p1_functional, run 3.	FALSIFIED
H3	Diagnostic; co-movement direction reported without falsification status	$\Delta_{gross} = -3.12pp$; $\Delta_{naive} = +2.50pp$. Decoupled directional change. Magnitude ratio 0.80.	DIAGNOSTIC

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.1). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

Gonzalez Castro, P. U. (2026). *Naive-Phantom Rate Longitudinal Stability: AI Presence Index v0.10 — Mint, designed-for-test extension*. Third System. thirdsystem.ai/v10-naivephantom-stability

Underlying datasets

v0.10 deposit. OSF project [ec6wh](https://osf.io/ec6wh/), path `/v10/`. Inherits v0.9 raw responses without modification (filter only). Adds: filtered Mint subset (`responses_mint.csv`), classification outputs (`classifications.csv`), canonical scoring (`canonical_scoring.{csv,json}`), E3 sensitivity scoring, `DEVIATIONS.md`, adjudication log, build scripts (`filter_v10.py`, `classify_v10.py`, `score_v10.py`, `build_charts_v10_naivephantom.py`, `build_report_v10.py`), and inherited classifier configuration (`registries/v07_caveat_classifier.json`).

Cross-cited deposits. v0.9 raw responses, v0.7 caveat-classifier specification, and v0.6 baseline data remain available in their respective OSF deposits, unchanged. v0.10 makes no modifications to upstream data; the naive-phantom rate is recovered from v0.9's deposited data by applying the v0.7 classifier. AIAS Presence Measurement Protocol v1.1 governs the program.

Source archive. The v0.10 build pipeline (chart generator, brand-format report builder, scoring scripts, classifier wrapper, content modules) is included in the deposit. The full pipeline is reproducible from raw v0.9 responses with a Python 3.14 environment, the brand JSON, and Akkurat Pro registered.

Methodology log: Pre-registration locked at git commit [8767f44](https://github.com/thirdsystem/thirdsystem/commit/8767f44) (tag **v0.10-prereg**) on 2026-05-09, prior to any application of the v0.7 caveat-classifier to the v0.9 Mint response subset. Four deviations from pre-reg descriptive language documented post-lock in `DEVIATIONS.md`. AIAS Presence Measurement Protocol v1.1.

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